

Python for QA

Readable programming language with a strong ecosystem for automation, APIs, data and utilities.

Why QA uses Python

Python is used for Web/API automation, scripts, test-data generation, data analysis and tool integration.

Installation

Download from the official site. On Windows, use the installer option to add Python to PATH when available. Validate with `python --version` and `pip --version`.

If PATH fails

Locate the Python installation and Scripts directory, add them to environment variables and reopen the terminal.

Virtual environment

Use `python -m venv .venv` to isolate dependencies. In PowerShell activate with `.\venv\Scripts\Activate.ps1`.

pip and requirements

pip installs packages. requirements.txt records reproducible dependencies; install with `pip install -r requirements.txt`.

Fundamentals

Learn variables, types, conditions, loops, functions, collections, modules, exceptions and OOP as needed.

QA in practice

Combine Python with Pytest, Playwright, requests and other libraries while keeping secrets outside code.

QA tip

■ QA Tip: programming for QA means understanding and maintaining automation, not only copying scripts.

Official resources

Site oficial: <https://www.python.org/downloads/>